

Eco-Friendly Chitosan Carboxymethyl Cellulose Composite for Enhanced Delivery of Nitrogen, Phosphorus, and Potassium in Sustainable Agricultural Systems

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Priya E, Jay Sharma, Ananya Chugh, Sudipta Sarkar,* and Pradip K. Maji*



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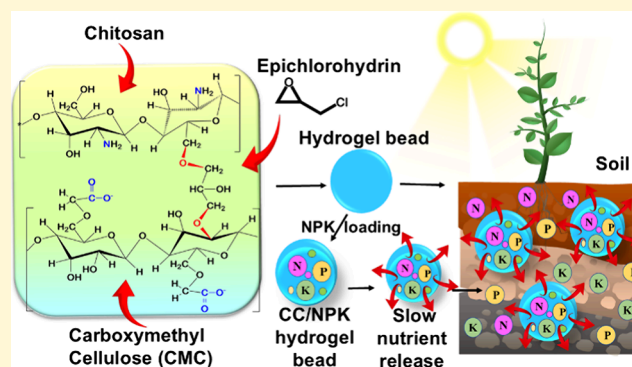
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ABSTRACT: Considering the vital requirement for efficient water and nutrient utilization to enhance crop production, it is necessary to implement effective strategies for managing these resources. NPK-loaded chitosan-carboxymethyl cellulose (CC/NPK) hydrogel beads were developed as an approach to enhance water and nutrient management. CC/NPK, a hydrogel with a biodegradable polymeric backbone impregnated with NPK, enhances plant growth through controlled nutrient delivery. For the CC/NPK hydrogel synthesis, CC beads were loaded with NPK, achieving a loading capacity of 94%. CC/NPK releases nutrients gradually over 20 days in water and has a water adsorption of about 38.5% in distilled water. After exposure to dry conditions, absorbed moisture was progressively released over 7 days. When placed in soil, over 80% of the hydrogel decomposed in 3 months via soil microbes. The plant study revealed that the CC/NPK hydrogel, having 25% of the conventional dose of NPK, exhibited better plant growth than conventional fertilizers.



Maintaining global food security is a major challenge, intensified by population growth, declining agricultural yields, and climate change.^{1,2} Inefficiencies in agrochemical delivery and plant usage (averaging 10–20%) exacerbate these difficulties. Therefore, low-cost, high-efficiency, sustainable agricultural solutions are essential.^{3–5} Enhanced-efficiency fertilizers minimize nutrient runoff, leaching, and volatilization while reducing environmental impacts such as eutrophication, biodiversity loss, and soil contamination.^{6,7} They also improve nutrient-use efficiency and crop productivity.⁸ Recently, agricultural practice has shifted toward biodegradable fertilizers to boost plant growth.⁹ Hydrogels developed from cross-linked polymer networks have become an exciting opportunity for modern agriculture. Their natural water adsorption property and polymer–water interaction are responsible for retaining and holding a large amount of water.¹⁰ Additionally, they address challenges related to conventional fertilizer, such as excessive nutrient availability for plants, which, instead of improving agrochemical delivery, often harms plants and the environment.¹¹ In addition to this issue, slow-release fertilizers (SRFs) made of nonbiodegradable polymers create another problem, releasing microplastics and degrading soil quality.^{12,13} This issue can be addressed using biodegradable SRFs made of materials like polysaccharides, which are abundant, cheap, sustainable, and biocompatible.¹⁴ Yet, four significant

challenges persist: cost-effectiveness with biodegradability, scalability and stability, overcoming slow desorption kinetics, and multifunctionality for water-use efficiency and to lessen adverse environmental effects.^{15,16}

Previous NPK slow-release formulations have often relied on more complex matrices. These systems typically introduce synthetic monomers, PVA, or inorganic fillers and frequently require multistep coating or curing procedures, which can increase material and processing costs and leave non-biodegradable residues in soil. By contrast, many systems still rely on nonbiodegradable or complex matrices, so fully biodegradable polysaccharide-based hydrogels without synthetic monomers or inorganic fillers that provide high nutrient loading, controlled release, and agronomic benefits at reduced fertilizer doses remain of particular interest. In this study, chitosan-carboxymethyl cellulose NPK (CC/NPK) hydrogel beads were investigated for sustainable agriculture. The hydrogel beads were synthesized by a green process, in

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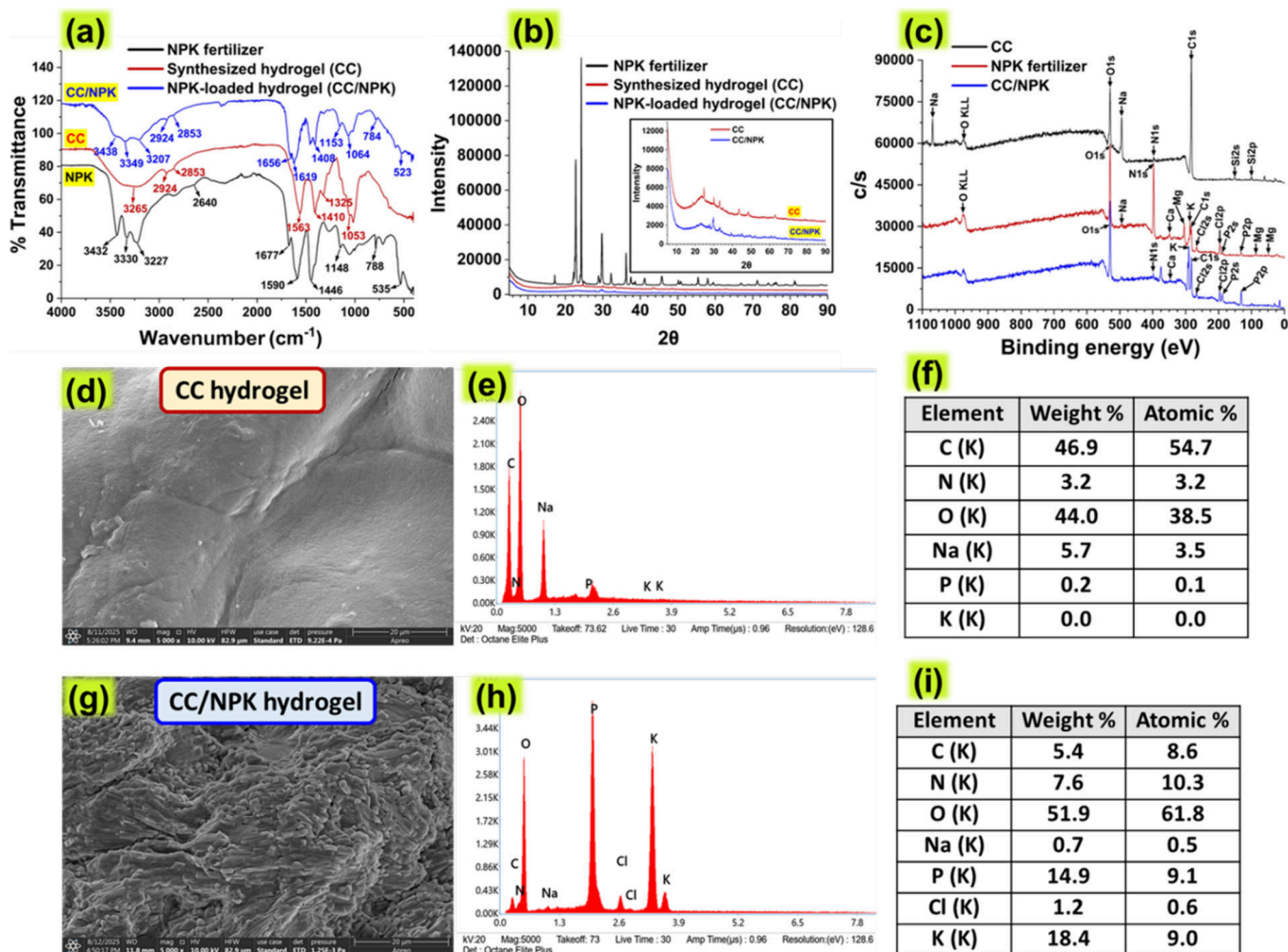


Figure 1. (a) FTIR spectra, (b) XRD patterns, and (c) wide scan XPS spectra of the synthesized CC hydrogel, NPK fertilizer, and CC/NPK hydrogel. FE-SEM/EDX analysis of the synthesized CC hydrogel with (d) FE-SEM micrograph, (e) elemental spectrum, and (f) element percentages. FE-SEM/EDX analysis of the CC/NPK hydrogel with (g) FE-SEM micrograph, (h) elemental spectrum, and (i) element percentages.

which chitosan and carboxymethyl cellulose (CMC) were cross-linked using a negligible amount of epichlorohydrin. The synthesis requires no initiator, catalyst, or multilayer coating and uses only aqueous processing of commercial chitosan, CMC, or NPK fertilizer, highlighting the simplicity of the CC/NPK hydrogel preparation. CMC was used as a hydrophilic filler in a hydrogel matrix to absorb water. When chitosan-carboxymethyl cellulose (CC) hydrogel beads are immersed in an NPK solution, they swell and absorb water, and nutrient ions are also entrapped in the porous polymeric network, resulting in CC/NPK hydrogel synthesis. Besides physical absorption, ionic functional groups (amino group in chitosan and carboxyl group in CMC) provide sites for electrostatic interaction for both cationic (e.g., potassium) and anionic (e.g., phosphate and nitrate) species. The CC/NPK hydrogel synthesis requires no postprocessing or energy-intensive purification. The combination of swelling and ionic interactions enhances effective nutrient loading and helps retain nutrient ions in the hydrogel matrix. Further, this enabled sustained nutrient release. Figure S1 presents a schematic of the synthesis process for the CC/NPK hydrogel beads, where under alkaline conditions the hydroxyl groups of CMC or chitosan undergo deprotonation, forming alkoxide anions (RO^-). Nucleophiles react with epichlorohydrin, displacing

chloride ions and forming ether bonds. The resultant epoxide intermediates facilitate subsequent attacks by alcoholate anions, leading to the cross-linking of chitosan and CMC during the synthesis of CC beads. This methodology allows for the preservation of carboxylic groups on CMC and amine groups on chitosan for the adsorption of both cationic and anionic nutrients, respectively, on the synthesized CC hydrogel. Material balance calculations conducted during multiple synthesis sessions revealed that the hydrogel prepared from chitosan and carboxymethyl cellulose (CC hydrogel) can load up to 0.94 g of NPK fertilizer per gram of hydrogel, thereby increasing the final weight of 1 g of CC hydrogel to 1.94 g of CC/NPK.

The FTIR spectra in Figure 1a confirm the successful incorporation of NPK into the CC hydrogel, showing characteristic chitosan/CMC bands (3265 cm^{-1} : $\text{O}-\text{H}/\text{N}-\text{H}$; 1053 cm^{-1} : $\text{C}-\text{O}-\text{C}$)⁷ combined with NPK fertilizer peaks ($3432/3330\text{ cm}^{-1}$: $\text{N}-\text{H}$ urea; 788 cm^{-1} : $\text{P}-\text{O}$; 1677 cm^{-1} : $\text{C}=\text{O}$ urea) with minimal peak shifts.¹⁷ A further detailed explanation of the FTIR results is provided in the Supporting Information (Section S2.3). The XRD pattern for NPK fertilizer revealed peaks related to urea, NaNO_3 , KCl , KH_2PO_4 , and KNO_3 , as well as CaCO_3 and SiO_2 as fillers with the NaCl impurity. XRD patterns of the CC and CC/

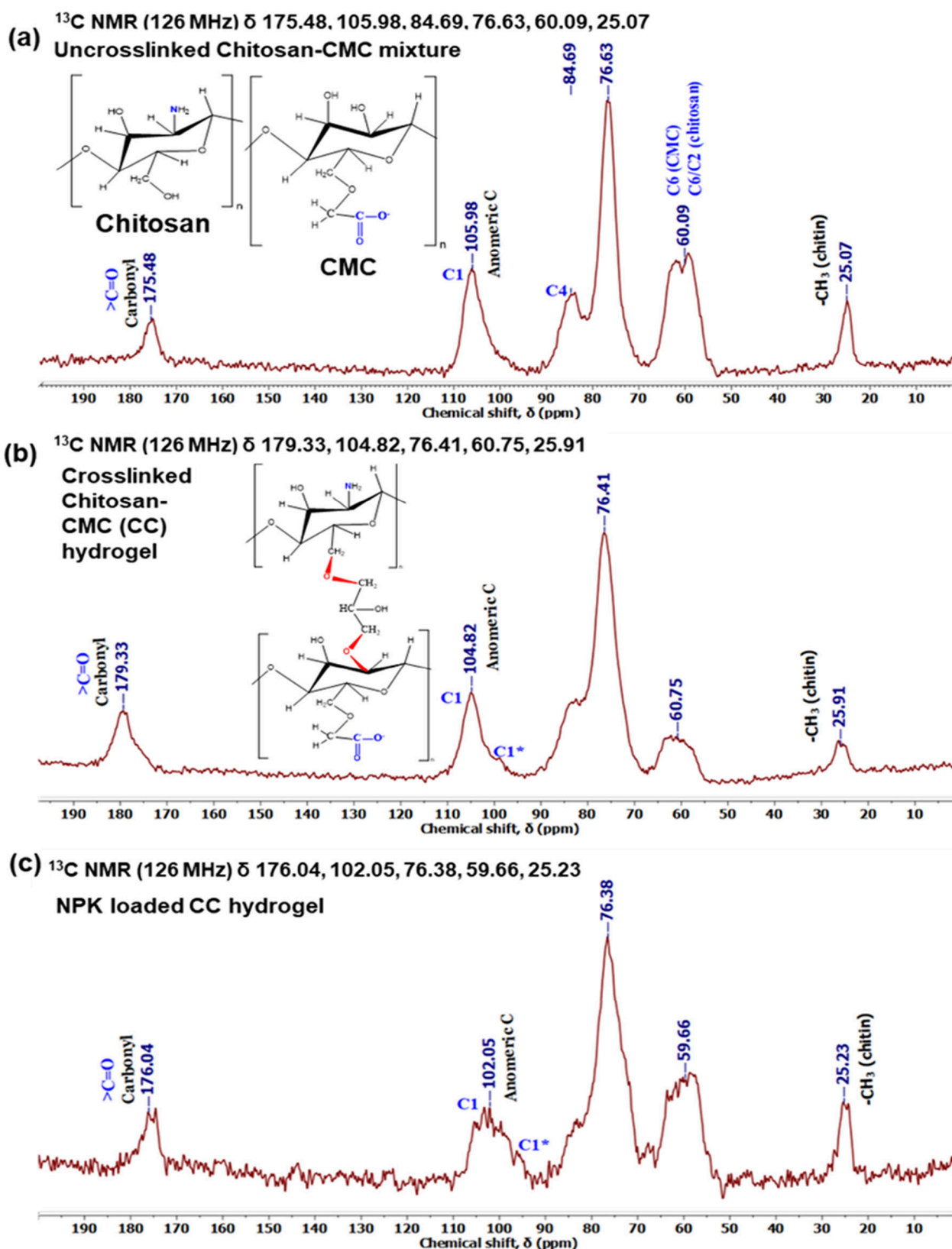


Figure 2. Solid-state ^{13}C NMR spectra of (a) un-cross-linked physical mixture of chitosan and CMC, (b) chitosan-CMC (CC) hydrogel cross-linked with epichlorohydrin, and (c) NPK-loaded CC hydrogel.

NPK showed predominantly amorphous behavior with minor crystalline peaks as opposed to NPK fertilizer (Figure 1b). The XRD pattern of CC/NPK (inset of Figure 1b) revealed some peaks with very low intensity, mainly indicative of SiO_2 ,

CaCO_3 , NaNO_3 , KNO_3 , K_2HPO_4 , NaH_2PO_4 , NaCl , and KCl . JCPDS file numbers of all of the compounds found in the XRD patterns are given in Table S1. Figure 1c shows the wide scan XPS spectra of the CC hydrogel, NPK fertilizer, and CC/NPK.

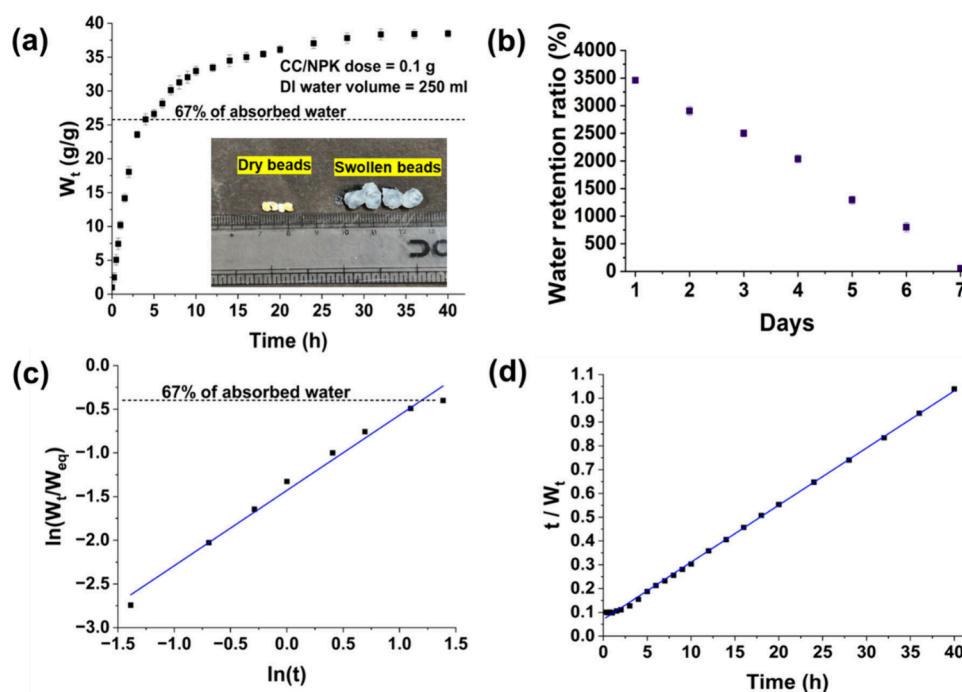


Figure 3. (a) Water absorption capacity of CC/NPK hydrogel beads in aqueous solution, (b) water retention characteristics of completely saturated CC/NPK beads, (c) swelling kinetics displaying a plot of $\ln(W_t/W_{eq})$ versus $\ln t$, and (d) swelling kinetics displaying a plot of t/W_t versus t .

CC does not show any P or K peaks, but a small peak of N is observed, which can be attributed to the presence of $-\text{NH}_2$ in chitosan. NPK fertilizer shows clear peaks of N, P, and K, along with peaks of Mg and Ca. CC/NPK shows clear, significant peaks of P and K, and a small N peak indicates a good incorporation of NPK fertilizer into the CC hydrogel. High-resolution deconvoluted XPS spectra of N 1s, P 2p, and K 2p peaks are presented in Figure S2 and discussed in the Supporting Information (Section S2.4).

FE-SEM/EDX analyses were performed to examine the surface morphology of the CC and CC/NPK hydrogels and to assess the NPK loading. Figure 1d suggests a smooth surface morphology of CC, typical of a hydrogel. Figure 1e,f shows the elemental spectrum and weight percentages of the major elements in the CC hydrogel beads, respectively, revealing almost no P or K, with a minor N content of 3.2% by weight, likely attributable to N in the chitosan structure. Figure 1g shows the FE-SEM image of the sliced cross-section of the CC/NPK hydrogel with visible surface morphology changes and the incorporation of NPK. Figure 1h,i presents the elemental spectrum and weight percentages of the predominant elements for the CC/NPK. Distinct high peaks of P, K, and O signify the sorption of K_2O and P_2O_5 onto the synthesized hydrogel, demonstrating sorption percentages of 14.9% and 18.4% by weight for P and K, respectively. A notable increase in the weight percentage of N (Figure 1i) in CC/NPK was also noted. The elemental mapping of the CC/NPK hydrogel is presented and discussed in Figure S3 (Section S2.5). Thermogravimetric analysis, depicted in Figure S4, demonstrated satisfactory thermal stability of the CC/NPK hydrogel up to 250 °C, with 50% weight loss and a total weight loss of 66.6% at 800 °C. Details on the degradation of the CC hydrogel are provided in the Supporting Information (Section S2.6).

Figure 2 presents the solid-state (SS) ^{13}C nuclear magnetic resonance (NMR) spectra of the un-cross-linked physical mixture of chitosan and CMC, cross-linked CC hydrogel, and CC/NPK hydrogel. Compared to the un-cross-linked mixture (Figure 2a), in the cross-linked spectrum (Figure 2b), the downfield shift of the carbonyl peak at 179.33 ppm,^{18,19} upfield shift of the anomeric carbon (C1) peak,^{20,21} disappearance of the 84.69 ppm peak, and appearance of a shoulder to the 76.41 ppm peak^{22,23} indicate successful cross-linking through ether linkages between the hydroxyl group of C6 ($-\text{CH}_2\text{OH}$) of chitosan and the hydroxyl group of C2/C3 of CMC.^{24,25} Further, the CC/NPK hydrogel spectrum (Figure 2c) indicated intact structural integrity and changes in peak shift, indicating the successful incorporation of the ionic species of NPK fertilizer.²⁶ Moreover, liquid-state ^1H NMR spectra (Figure S5a,b) for the un-cross-linked mixture and cross-linked CC hydrogel also confirm effective cross-linking through ether linkages and preservation of the CMC and chitosan backbone structures. Further details of solid-state ^{13}C NMR and liquid-state ^1H NMR are provided in the Supporting Information (Section S2.7). Possible network formations are provided in Figure S6.

Hydrophilic CC/NPK hydrogel beads have $-\text{OH}$ groups, amine groups ($-\text{NH}_2$), and ionizable carboxylate ($-\text{COOH}$) groups. In polar solvents, such as aqueous solutions, these groups absorb water and cause the material to swell. The water absorption capacity of CC/NPK in deionized water at neutral pH over time is shown in Figure 3a. Maximum water absorption was 38.5 g/g, which was 38.5 times the dry CC/NPK weight. Figure 3b depicts the water retention of the fully swollen CC/NPK hydrogel. A linear decrease in the amount of sorbed water after 7 days was observed under room conditions (25 °C and 60% RH). However, higher RH and lower temperature are generally found in the soil matrix, enabling the CC/NPK hydrogel to retain water longer than under the lab

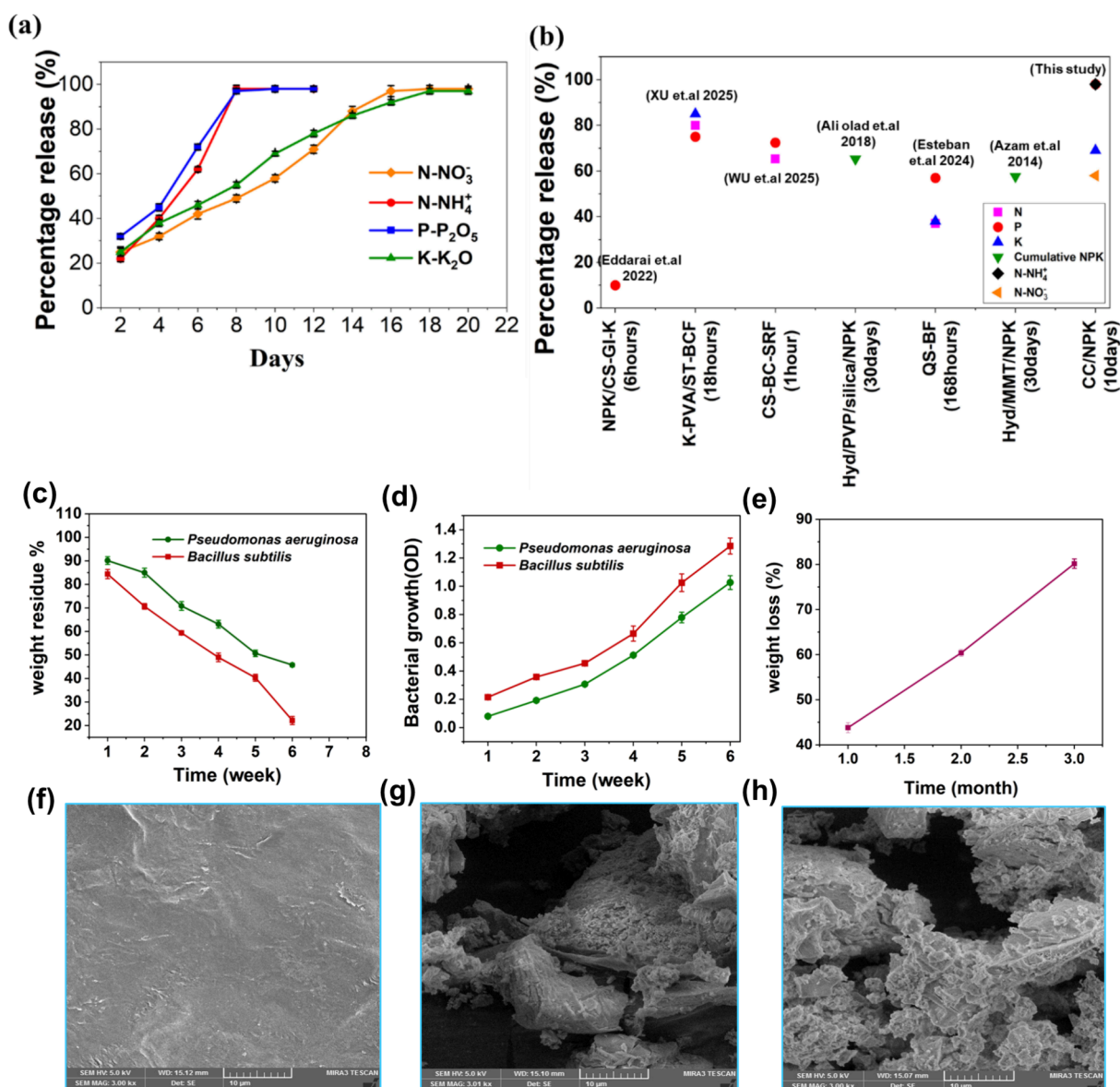


Figure 4. (a) Slow release of NPK from CC/NPK hydrogel beads in aqueous solution. (b) Comparative nutrient release behavior of CC/NPK hydrogel beads with other reported slow-release fertilizers (chitosan-kaolinite, PVA/starch-kaolinite coated biochar, metal-modified biochars, CMC nanocomposites, biochar-based NPK systems, and NaAlg-g-poly(AA-co-AAm)/MMT), which employ clay, biochar, or synthetic polymers, whereas the present system uses a fully biodegradable chitosan-CMC matrix cross-linked with a negligible amount of epichlorohydrin.^{15,27–31} (c) Weight loss profiles of CC/NPK hydrogel beads due to bacterial degradation. (d) Bacterial growth curves of *P. aeruginosa* and *B. subtilis* on CC/NPK hydrogel beads. (e) Weight reduction of CC/NPK hydrogel beads after three months, assessed by the soil burial biodegradation method. SEM images of (f) corresponding controls of CC/NPK hydrogel beads and (g, h) CC/NPK hydrogel beads following degradation by *B. subtilis* and *P. aeruginosa* after 6 weeks, respectively.

environmental conditions. Thus, the CC/NPK hydrogel gently releases water, minimizing root-zone water shortages in dry and semiarid regions and enabling sustainable agriculture. Understanding agricultural water transport requires investigating the water diffusion mechanism and swelling kinetics of CC/NPK hydrogel beads (Supporting Information, Section S1.8). Figure 3c presents the diffusion kinetics of the CC/NPK hydrogel, and the linear fitting showed a correlation coefficient of 0.98, indicating a good match. In the fitted model, n (indicative of the diffusion type) and k (related to the structural network of the hydrogel) were 0.862 and 0.245, respectively. Further, Table S2 demonstrates the parameters of swelling and diffusion kinetics. When $n < 0.5$, water molecule transport in the hydrogel pore network is dominated by

Fickian diffusion, while for n values between 0.5 and 1, non-Fickian transport is predominant. Most non-Fickian diffusion involves a dual mechanism. Initially, water is rapidly absorbed as water molecules enter the free volume of the polymer. However, water absorption is also hindered by polymer relaxation, structural rearrangement, and swelling due to the connections of the polymer chains between hydrophilic groups and water. When the polymer chains expand and relax, they impede the mobility of water molecules upon contact. Since polymer relaxation is linked to diffusion, this leads to anomalous diffusion behavior. The calculated value of $n = 0.862$ indicates that the swelling of the CC/NPK hydrogel beads adheres to a non-Fickian (anomalous) diffusion process. Figure 3d provides the pseudo-second-order model, which

plots t/W_t versus t using experimental data to determine kinetic parameters. Although the power-law model used above provides details about the diffusion mechanism, the kinetic modeling further clarifies the swelling process. A linear curve ($R^2 = 0.99$) confirmed the pseudo-second-order swelling kinetics. The theoretical W_{eq} of the CC/NPK hydrogel was 41.5 g/g, similar to the experimentally obtained equilibrium swelling capacity. The close agreement between the experimental and theoretical equilibrium swelling capacity validates the reliability of the applied pseudo-second-order model.

As observed from Figure 4a, the release of nutrients was as follows: 25%, 22%, 32%, and 25% on day 2; 32%, 40%, 45%, and 38% on day 4; 49%, 98%, 97%, and 55% on day 8; and 71%, 98%, 98%, and 78% on day 12 for $N-NO_3^-$, $N-NH_4^+$, $P-P_2O_5$, and $K-K_2O$, respectively. By day 20, $N-NH_4^+$, $P-P_2O_5$ were entirely released, while $N-NO_3^-$ and $K-K_2O$ attained 98% release, respectively. Nitrate-N has a good slow-release rate, primarily due to its electrostatic interaction with $-NH_3^+$ of the hydrogel matrix, making release dependent on polymer relaxation. At the neutral pH of the release medium, phosphate exists in two forms: $H_2PO_4^-$ and HPO_4^- , which remain strongly hydrated, making it less likely to stay bound and form an ionic interaction like $-NO_3^-$. Therefore, we observed a moderate phosphate release rate, reaching 100% after 10 days. Ammonium-N also expresses a limited slow-release rate because of its interaction with the deprotonated ($-COO^-$) group of the matrix and could be easily displaced by protons due to the acid–base exchange ($-COO^- \cdot NH_4^+ + H_2O \leftrightarrow COOH + NH_4^+$). Another factor causing its release is co-ion exclusion with the protonated amino group ($-NH_3^+$) of the matrix. Lastly, an excellent slow release is observed with potassium. K^+ interaction is much more stable with $-COO^-$ than NH_4^+ , and it is not easily displaced. Thus, we observed a strong, sustained release over 20 days. As discussed earlier, the swelling kinetics in water confirmed a non-Fickian mechanism; the staggered release pattern of NPK here is also consistent with the occurrence of both diffusion and polymer relaxation for the transport of the nutrients. The chitosan-CMC hydrogel contains protonated amino groups of chitosan $-NH_3^+ \leftrightarrow NH_2 + H^+$ ($pK_a \approx 6-6.5$) and deprotonated carboxyl groups of CMC ($-COO^-$) at neutral pH of the release medium, which create charged microdomains within the hydrogel matrix. Consequently, the nutrient release also depends on the electrostatic interactions of the matrix. These results demonstrate effective slow-release behavior, enabling sustained nutrient delivery, improved plant uptake, and reduced leaching losses compared with conventional fertilizers. Compared with other slow-release fertilizers (Figure 4b), the CC/NPK hydrogel exhibited a more sustained release over 20 days. In addition to this favorable release, the CC hydrogel matrix attains an NPK loading of 0.94 g/g and a maximum water uptake of 38.5 g/g while sustaining $N-NO_3^-$ and $K-K_2O$ release up to 20 days and releasing $N-NH_4^+$ and PP_2O_5 almost completely within 8–12 days. As summarized in Figure 4b, the overall release duration and staggered profile of CC/NPK are comparable to or better than those reported for chitosan-kaolinite, PVA/starch-kaolinite coated biochar, metal-modified biochars, CMC nanocomposites, biochar-based NPK systems, and NaAlg-g-poly(AA-co-AAm)/MMT.^{13,27–31} These studies rely on clay, biochar, or synthetic polymers, whereas CC/NPK achieves similar performance using a chitosan-CMC matrix with a negligible amount of epichlorohydrin as a cross-linker.

The biodegradation capability of CC/NPK hydrogel beads was assessed by the McFarland turbidity test utilizing two bacterial strains: *Pseudomonas aeruginosa* (Gram-negative) and *Bacillus subtilis* (Gram-positive). The percentage of weight loss of the CC/NPK hydrogel beads over time was measured to determine microbial degradation. The progressive weight reduction of CC/NPK hydrogel beads was quantified over time in a bacterial culture (Figure 4c). Figure 4d shows that the optical density of the bacterial growth curve rose over time. Active microbial growth (Figure 4c,d) and structural breakdown (Figure 4f–h) confirm biodegradation, with sterile controls showing no degradation. The results confirm microbial-driven biodegradation. CC/NPK hydrogel beads were gradually released during the process and were consumed by microorganisms for their growth. Field-emission scanning electron microscopy (FE-SEM) revealed morphological and structural damage during bacterial degradation. Figure 4f–h presents CC/NPK hydrogel beads obtained from both sterile and bacterial-inoculated media. The results reveal that hydrogel beads exposed to the bacterial cultures *Bacillus subtilis* and *Pseudomonas aeruginosa* exhibited an evident structural breakdown. In contrast, beads kept under sterile conditions without bacteria preserved their original structural morphology. Moreover, these findings underscore the robust biodegradation potential of the chosen bacterial species and highlight the biodegradability of CC/NPK hydrogel beads in microbial environments. The CC/NPK hydrogel beads were buried in soil for different intervals to assess degradation in actual conditions. The weight loss of the beads was measured at various intervals, as illustrated in Figure 4e. The findings demonstrate a significant deterioration rate, with around 43% of the beads decomposed after 30 days and over 80% after 90 days of soil burial. The soil degradation study demonstrates significant disintegration under natural environmental circumstances, facilitating a continuous release of organic carbon and nutrients into the soil. CC/NPK hydrogel beads show strong potential as slow-release fertilizers. Their capacity to degrade under microbiological conditions and gradually release nutrients indicates promising application in sustainable agriculture.

The beneficial effects of the slow-release fertilizer properties of CC/NPK hydrogel beads were evaluated in a controlled pot experiment. Pot experiments (5 groups, 3 replicates, Figure S7) compared control, conventional NPK, and CC/NPK at 25/50/100% equivalent NPK doses. All relevant calculations are provided in the Supporting Information (Section S2.9). In this study, 25% CC/NPK hydrogel treatment improved plant development more than conventional NPK. The entire dose of conventional NPK in a 2 kg pot generates a concentrated salt environment. This causes high ionic strength, which hinders seed water absorption (osmotic stress) and causes poor growth. The plant needs only a modest amount of nutrients initially; therefore, extra fertilizer is harmful. In contrast, the 25% CC/NPK treatment provided a more conducive environment for plant growth by optimizing nutrient absorption. Interestingly, for the hydrogel treatment, plant growth followed the order 25% > 50% > 100%. This suggests that while the hydrogel helped buffer nutrient release, higher concentrations (50% and 100%) led to localized nutrient levels that exceeded the needs of early seedlings. However, the soil mass in an agricultural field is way bigger than that in a pot, and a higher percentage of the nutrients would be lost via soil fixing, leaching, and volatilization compared to the pot. The crops

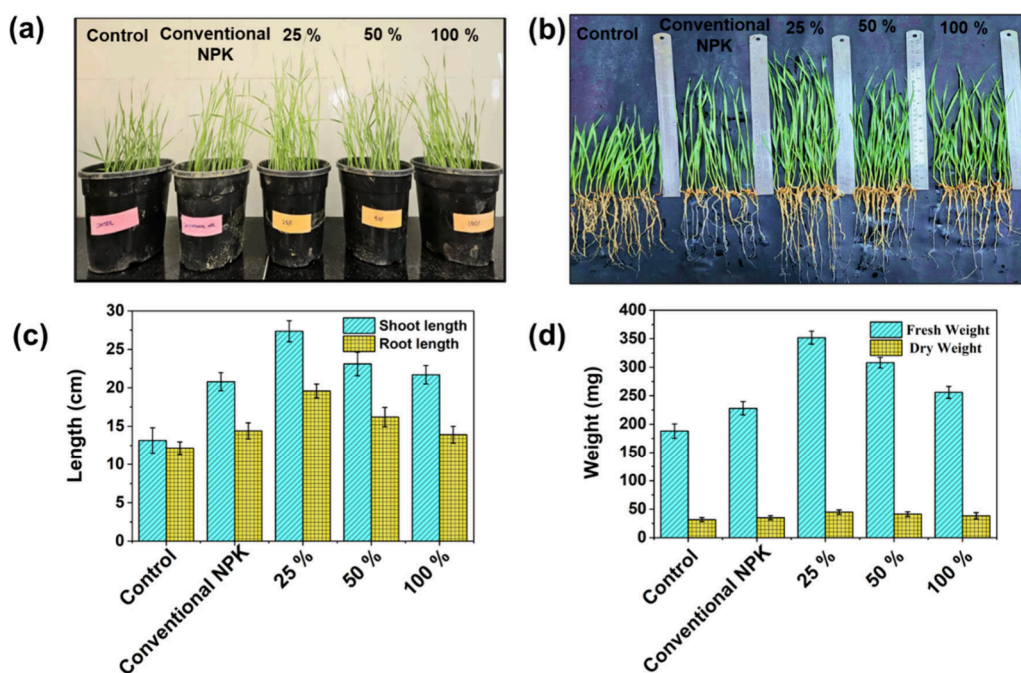


Figure 5. (a, b) Plants and their shoots and roots after 2 weeks of seeding. The plants received varying amounts of the CC/NPK hydrogel beads and conventional NPK (20:20:20) fertilizer. Measurements included (c) shoot and root lengths and (d) fresh and dry weights of the plants after 2 weeks, assessed based on the different treatment groups.

would eventually need more fertilizers in the field. Consequently, farmers would use a greater quantity of NPK than optimally required. The superior performance of the 25% CC/NPK hydrogel treatment demonstrates the benefits of controlled-release systems. The CC/NPK hydrogel gradually released the nutrients and maintained a favorable micro-environment for seed germination and early growth. 25% CC/NPK matched the nutrient requirements of the plant, reducing fertilizer input. This suggests that these CC/NPK hydrogels would help minimize fertilizer consumption while maintaining crop development, germination, and soil health. The data from the potted plants were evaluated to determine root and shoot lengths and their fresh and dry weights. Figure 5a presents pots following 14 days of plant growth. The findings demonstrate that the pot with 25% CC/NPK content exhibited a growth rate better than that of conventional NPK fertilizer. Figure 5b illustrates the root and shoot lengths of randomly selected plants. Figure 5c,d illustrates root and shoot lengths and fresh and dry weights, respectively. These data support the conclusion that plants receiving CC/NPK fertilizer had a higher growth rate than those receiving conventional fertilizer. A consolidated comparative discussion highlighting the simplicity of the synthesis process, the high nutrient loading with controlled-release behavior, and the demonstrated fertilizer dose reduction in pot trials is provided in the Supporting Information (Section S3).

In summary, beyond these mechanistic insights, CC/NPK is a fully biodegradable chitosan-CMC hydrogel carrier synthesized without synthetic monomers, PVA, or inorganic fillers and using only a trace amount of epichlorohydrin for cross-linking. The hydrogel achieves 94 wt % NPK loading, exhibits a 20 day staggered release of nitrate, ammonium, phosphate, and potassium, and absorbs up to 38.5 g/g of water while undergoing more than 80% biodegradation in soil within 90 days. Pot experiments demonstrate that supplying only 25% of the conventional NPK dose via CC/NPK yields better early

plant growth than 100% conventional NPK, highlighting enhanced nutrient-use efficiency and the potential to reduce fertilizer input in sustainable agricultural systems.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsmaterialslett.6c00100>.

Additional experimental details, materials, and methods, as well as supplementary figures and tables, including a schematic of the synthesis process, XRD reference data (JCPDS), FTIR spectra, high-resolution XPS spectra, EDX elemental mapping, TGA curves, ^{13}C and ^1H NMR spectra, and plant study calculations and results (PDF)

■ AUTHOR INFORMATION

Corresponding Authors

Sudipta Sarkar – Centre for Nanotechnology, Indian Institute of Technology Roorkee, Roorkee 247667 Uttarakhand, India; Department of Civil Engineering, Indian Institute of Technology Roorkee, Roorkee 247667 Uttarakhand, India; Phone: +91-1332284756; Email: sudipta.sarkar@ce.iitr.ac.in

Pradip K. Maji – Department of Polymer & Process Engineering, Indian Institute of Technology Roorkee, Saharanpur Campus, Saharanpur 247001 Uttar Pradesh, India; orcid.org/0000-0001-8112-554X; Phone: +91-7895965010; Email: pradip@pe.iitr.ac.in

Authors

Priya E – Centre for Nanotechnology, Indian Institute of Technology Roorkee, Roorkee 247667 Uttarakhand, India; orcid.org/0009-0001-3185-4792

Jay Sharma – Department of Civil Engineering, Indian Institute of Technology Roorkee, Roorkee 247667 Uttarakhand, India; orcid.org/0000-0002-9839-3302
 Ananya Chugh – Centre for Nanotechnology, Indian Institute of Technology Roorkee, Roorkee 247667 Uttarakhand, India

Complete contact information is available at:

<https://pubs.acs.org/10.1021/acsmaterialslett.6c00100>

Author Contributions

CRedit: Priya E conceptualization, data curation, formal analysis, investigation, methodology, project administration, resources, software, validation, visualization, writing - original draft, writing - review & editing; Jay Sharma data curation, formal analysis, investigation, methodology, software, validation, writing - original draft, writing - review & editing; Ananya Chugh data curation, formal analysis, investigation, methodology, resources, software, writing - original draft, writing - review & editing; Sudipta Sarkar conceptualization, formal analysis, funding acquisition, methodology, resources, supervision, validation, visualization, writing - original draft, writing - review & editing; Pradip K. Maji conceptualization, formal analysis, funding acquisition, investigation, resources, supervision, validation, writing - original draft, writing - review & editing.

Notes

The authors declare no competing financial interest.

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